

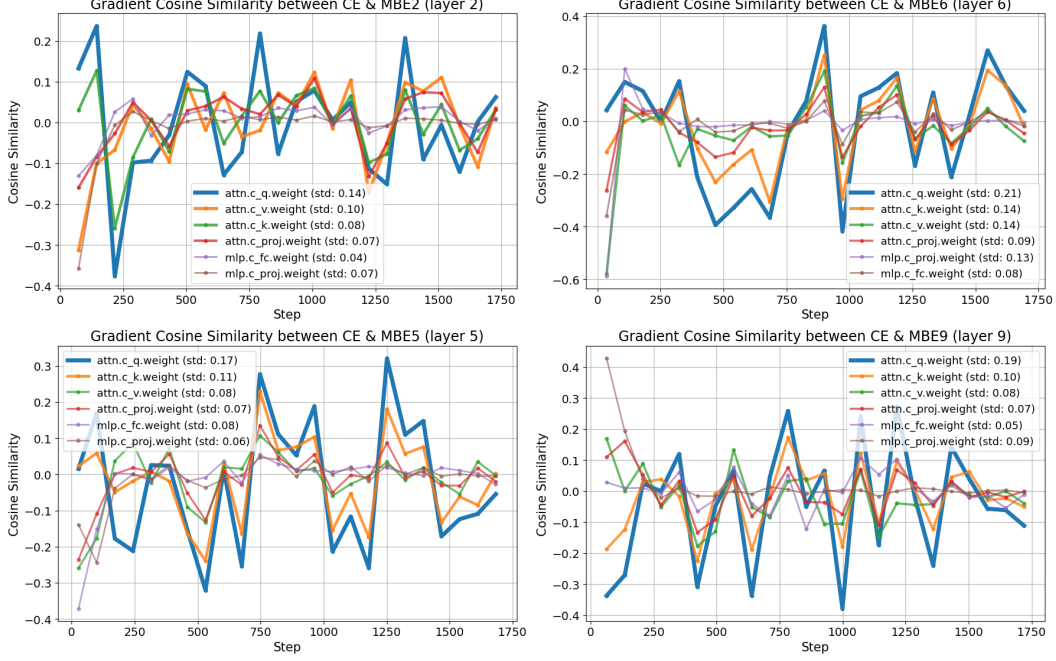


Figure 3: Cosine similarity between CE and MBE gradients over training. Alternating positive and negative phases indicate emergent memorization–compression cycles.

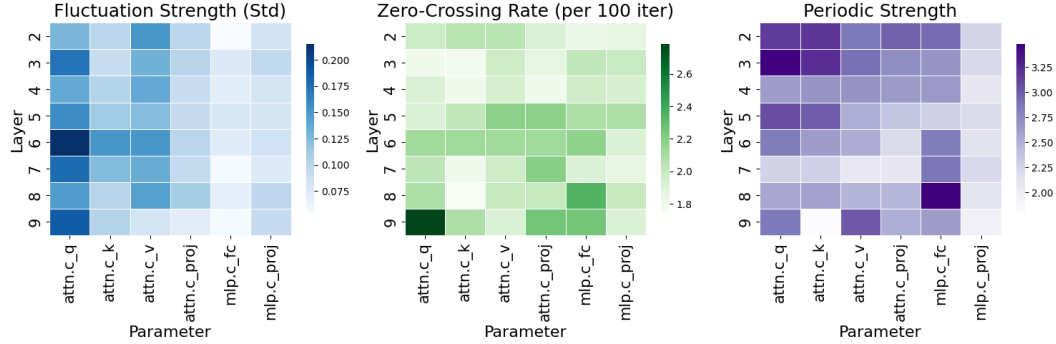
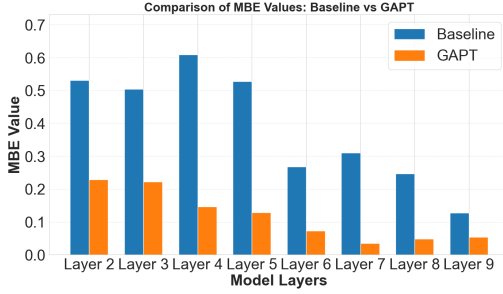


Figure 4: Oscillation metrics between CE and MBE gradients across layers and parameter groups. Left: standard deviation; center: zero-crossing rate; right: periodic strength (peak-to-mean PSD ratio).

constrained optimization objective in IBLM (Equation 2), and can exceed baseline training with cross-entropy target alone.

4.3 Conflicting memory resolution

Inspired by [18], which showed that sleep consolidation helps resolve memory conflicts, we design a synthetic experiment to test whether GAPT can mitigate representation interference between conflicting experiences. We use a 2-layer MLP f_θ with randomly initialized parameters and define a symmetric shift $\Delta\theta$. Inputs $X_1, X_2 \in \mathbb{R}^{10}$ are sampled from Gaussians with shared variance but distinct means:



Layer	Base	GAPT	Reduction
2	0.5313	0.2285	-56.99%
3	0.5039	0.2217	-56.01%
4	0.6094	0.1465	-75.96%
5	0.5273	0.1279	-75.74%
6	0.2676	0.0728	-72.81%
7	0.3105	0.0344	-88.91%
8	0.2471	0.0483	-80.44%
9	0.1270	0.0542	-57.32%
Avg	-	-	-70.52%

Figure 5: Left: Layer-wise MBE for baseline vs. GAPT. Right: Per-layer MBE reduction with GAPT.

$$X_1 \sim \mathcal{N}([1, 1, 1, 1, 1, 0, 0, 0, 0], \sigma^2 I),$$

$$X_2 \sim \mathcal{N}([0, 0, 0, 0, 0, 1, 1, 1, 1], \sigma^2 I)$$

The targets are defined as $Y_1 = f_{\theta+\Delta\theta}(X_1)$ and $Y_2 = f_{\theta-\Delta\theta}(X_2)$, producing two tasks with negatively aligned gradients. We compare GAPT against four baselines: single-task training, ordered training, mixed training, and GAPT applied to mixed batches with MBE regularization.

Table 2: Performance on conflicting experience learning. Lower L1/MBE and higher separation indicate better generalization and disentanglement.

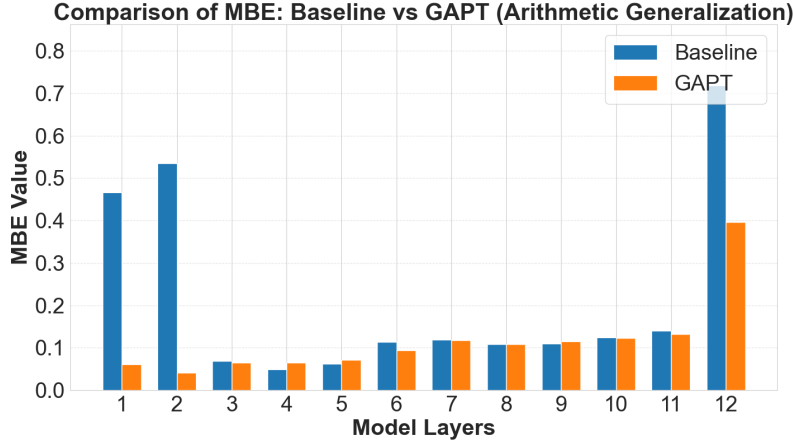
Strategy	L1 (Pos)	L1 (Neg)	MBE (Pos)	MBE (Neg)	Dist.	Sep. Ratio
Pos-only	0.02	0.71	0.18	0.45	-	-
Neg-only	0.92	0.02	0.36	0.36	-	-
Pos $\rightarrow$ Neg	0.43	0.04	0.19	0.15	2.43	2.84
Neg $\rightarrow$ Pos	0.02	0.57	0.19	0.43	1.86	2.33
Mixed	0.03	0.03	0.10	0.22	3.66	4.11
GAPT + MBE (Ours)	0.03	0.03	0.02 (-80%)	0.02 (-91%)	6.64 (+81%)	8.08 (+97%)

Table 2 summarizes results. Ordered learning, where the model is trained on one experience and then on the other, suffers from catastrophic forgetting: performance on the first task degrades significantly after exposure to the second. Mixed training alleviates this issue, achieving low L1 loss on both tasks and moderate representation separation. However, GAPT improves over mixed training in both respects: it maintains the same L1 accuracy while achieving a 97% increase in separation ratio and a 91% reduction in MBE.

These results suggest that the compression encouraged by GAPT not only preserves generalization performance but also promotes the disentanglement of conflicting memories. Moreover, compression and separation emerge in tandem during training, closely resembling the consolidation behavior observed in biological neural systems during sleep. This supports the view that compression serves not only as a generalization mechanism, but also as a functional tool for resolving interference in memory [18].

4.4 Arithmetic generalization

To evaluate whether GAPT improves generalization in pre-training language models, we conduct a controlled experiment using a synthetic arithmetic dataset. We pre-train a GPT-2 model from scratch to perform integer multiplication. The training dataset contains 10 million multiplication equations between integers with 1–3 digits. For evaluation, we prepare two test sets: an in-domain (ID) set with 10,000 examples also from the 1–3 digit range, and an out-of-domain (OOD) set with 10,000 examples involving 4–6 digit multiplications. An additional OOD validation set with 1,000 examples is used for early stopping.



Metric	Baseline (1085)	GAPT (Ours, 898)	Change
Entropy (ID)	0.010	0.011	+10%
Entropy (OOD)	4.334	2.817	-35%
Avg MBE (0–11)	0.218	0.115	-47%

Figure 6: Up: Comparison of baseline vs. GAPT on arithmetic generalization. Bottom: Arithmetic generalization performance summary. GAPT improves OOD generalization and yields more compact representations.

We tokenize the input and output sequences at the per-digit level and train the model for 1,750 iterations with a batch size of 16. Due to observed instability in OOD entropy, we adopt an early stopping strategy that halts training if validation loss increases by more than 20% after iteration 800.

As shown in Figure 6, GAPT improves generalization substantially. It reduces OOD entropy by 35% and average representation entropy (MBE) by 47%, while maintaining similar performance on the in-domain set. This supports our theoretical prediction that minimizing representation entropy leads to stronger generalization under the Information Bottleneck Language Modeling (IBLM) framework.

Interestingly, while MBE regularization is only applied to a subset of layers, we observe that GAPT achieves lower MBE even in unregularized layers (e.g., layers 0, 1, and 11), suggesting a degree of entropy compression generalization across the network. Notably, MBE in layer 1 is reduced by 92%, and in layer 11 by 45%.

5 Relevant work

The **Information Bottleneck (IB)** method was introduced in [2] to formalize the goal of retaining only task-relevant information in representations by minimizing entropy. A theoretical connection between generalization and the cardinality of discrete inputs was established in [6], and later applied to deep networks in [8]. However, the entropy-generalization link remained incomplete due to the gap between cardinality and entropy measures. Empirical evidence of a two-phase learning dynamic—early memorization followed by entropy compression—was presented in [15]. To quantify entropy in neural networks, matrix-based entropy (MBE) was proposed in [5], and later applied to LLMs in [14], where it correlated with embedding quality and revealed compression trends across checkpoints.

LLMs such as GPT-2 [11] generalize across tasks by scaling training data and parameters [4], effectively compressing the training corpus [17, 13]. While post-training methods like instruction tuning [9] improve usability, LLMs still struggle on out-of-distribution tasks such as reverse reasoning [22, 20] and multi-hop inference [20]. Recent advancements in RL with verifiable rewards (RLVR) have improved mathematical and coding performance [7], though often by narrowing base model behavior [10]. Finally, curriculum-based tokenization growth has been combined with pre-training to improve compression [21].